

θ 4–8 Hz · H-module

γ 30–100 Hz · L-module

Reimplementing the Hierarchical Reasoning Model

Brain-inspired latent reasoning capable of impressive accuracy on puzzle performance with modest training set sizes

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REFS: Wang et al., arXiv:2506.21734 (2025)
D. Kahneman, *Thinking, Fast and Slow* (2011)

01 Motivation

Problem: Most models are bad at solving complicated problems

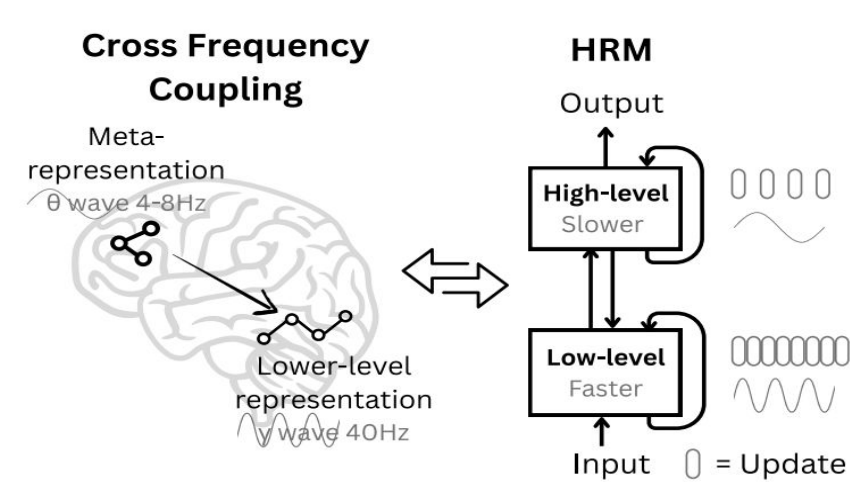
Reasoning should occur in latent space and not scale with output length

Proposed Solution: Stack transformer blocks in a way that mimics the brain

Lower frequency modules over long timescales (System 2)

Higher frequency modules for immediate details (System 1)

We aim to see if this proposed model performs well on Sudoku puzzles

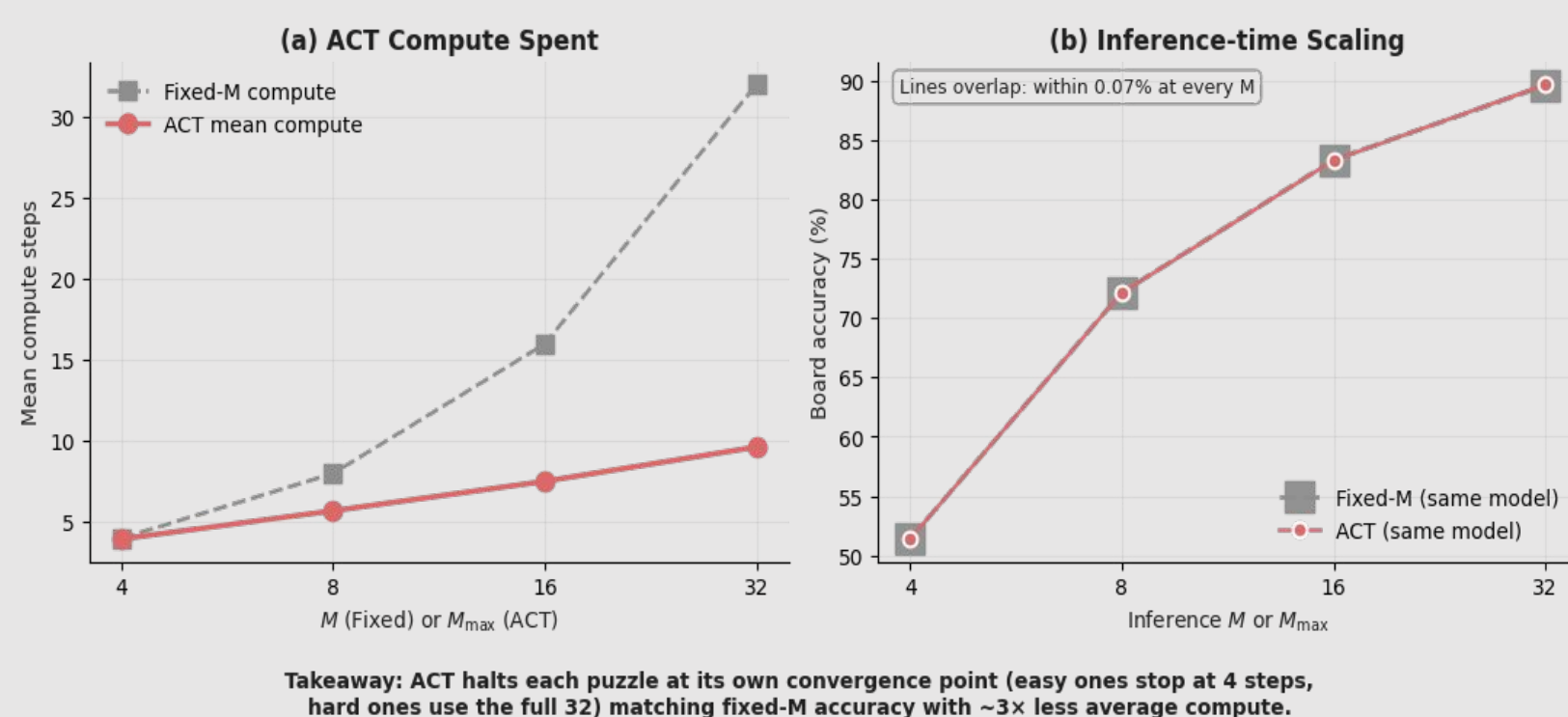
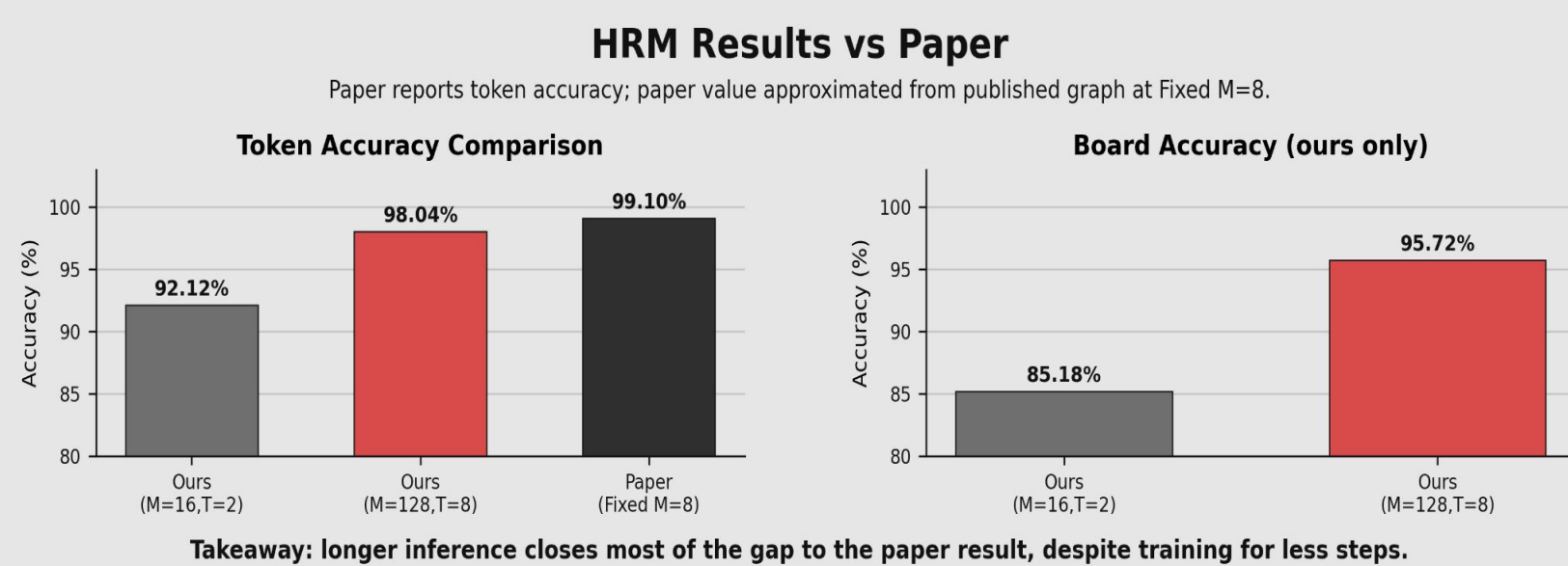


Benefits of the HRM:	Detriments of the HRM:
- Scalable post training - Empirically outperforms similarly sized models - Requires less training data than a BERT-style transformer - Increases effective model depth	- Not generative - Gradients are approximate - Inference time scales linearly with depth

03 Methods, Experiments & Results

Did we match the paper's results?

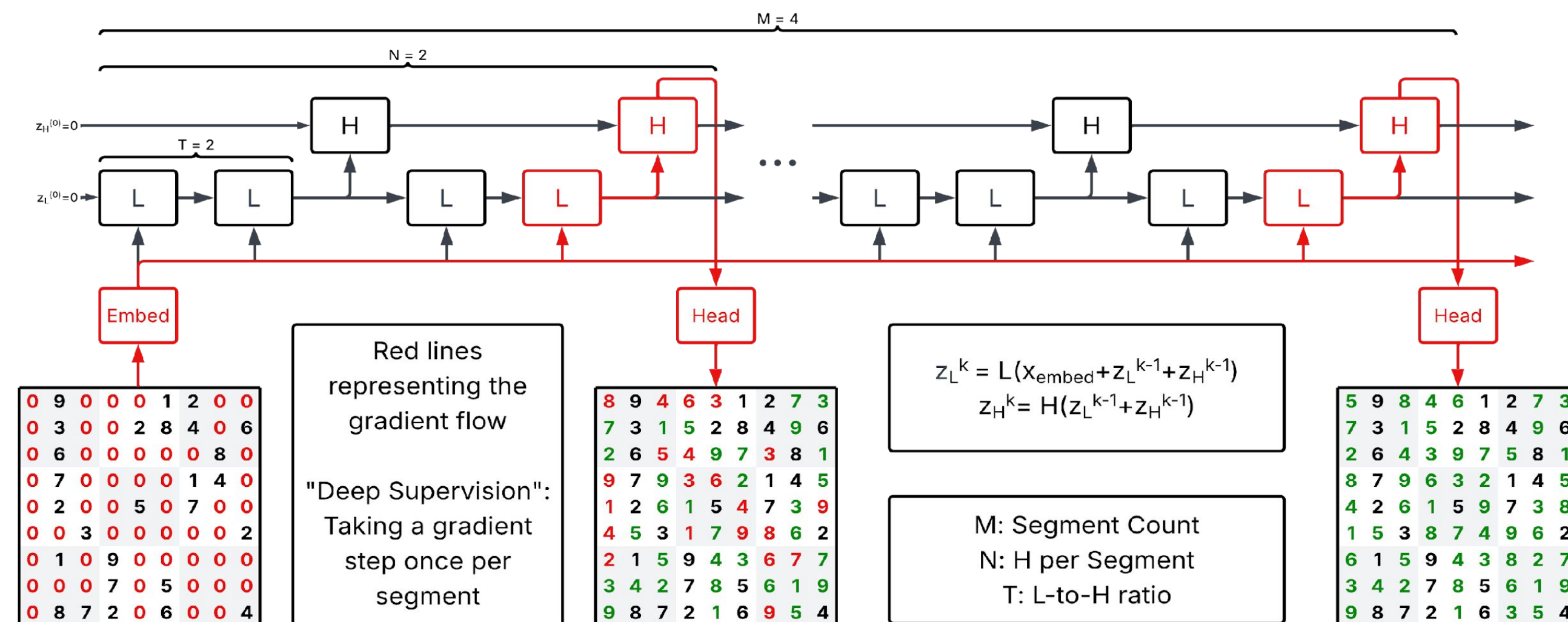
Methods and Primary Results Replication



Stats for Nerds:

- HRM-checkpoint-sudoku-extreme dataset (3M puzzle examples)
- Trained on 2¹⁸ examples, 33.6M parameters, 20% dropout, 328K steps
- Learning rate 1e-4, 5% linear warmup, cosine decay to 1e-5

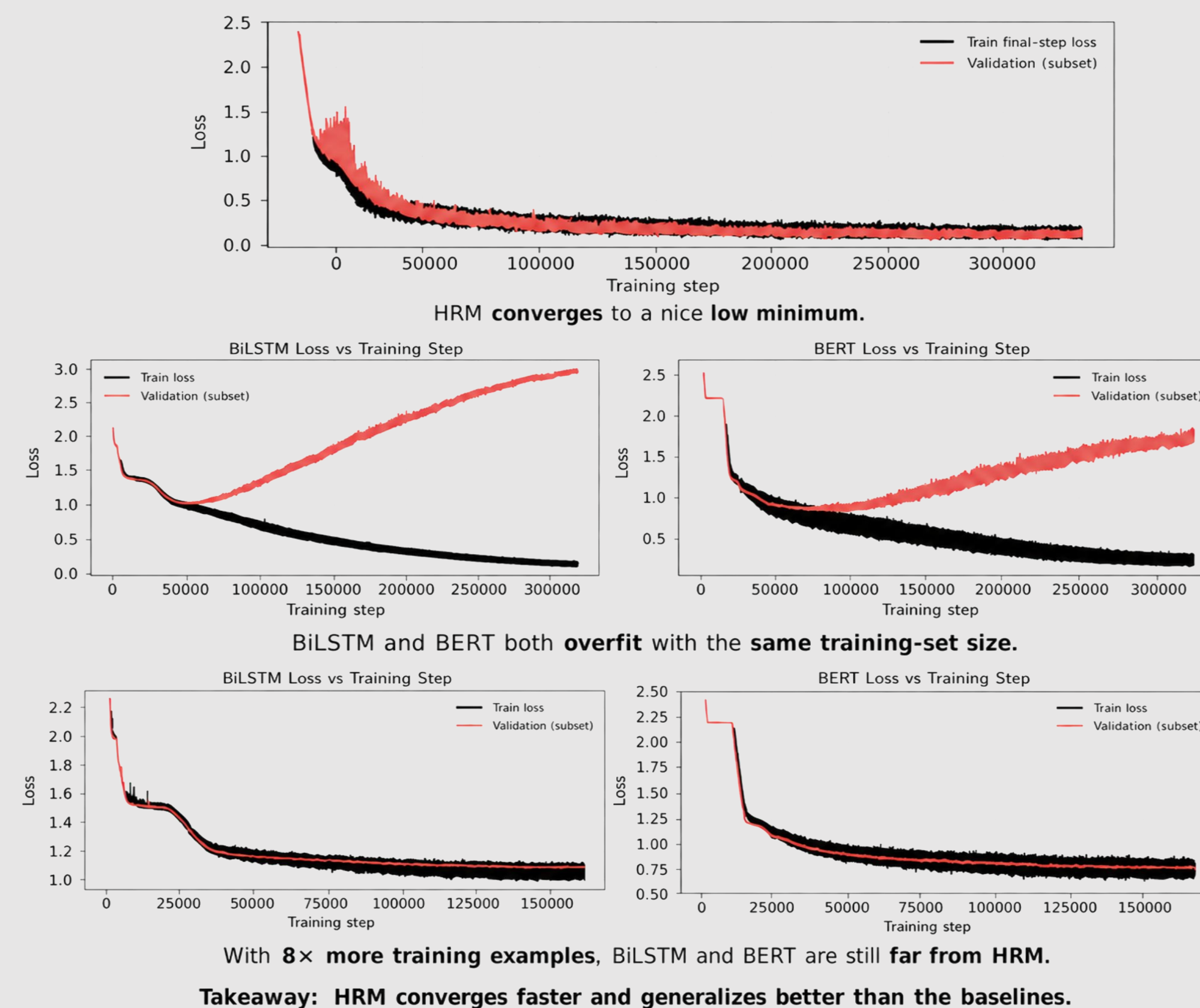
02 Architecture



A transformer called the **L-Module** processes frequently, while the **H-Module** processes less frequently. All **L-Modules** have the same weights, and so do all **H-Modules**. All **segments** are therefore identical, so during training, gradient steps happen **once per segment**, which is referred to as **Deep Supervision**. To make this feasible, a **one-step gradient** approximation is used.

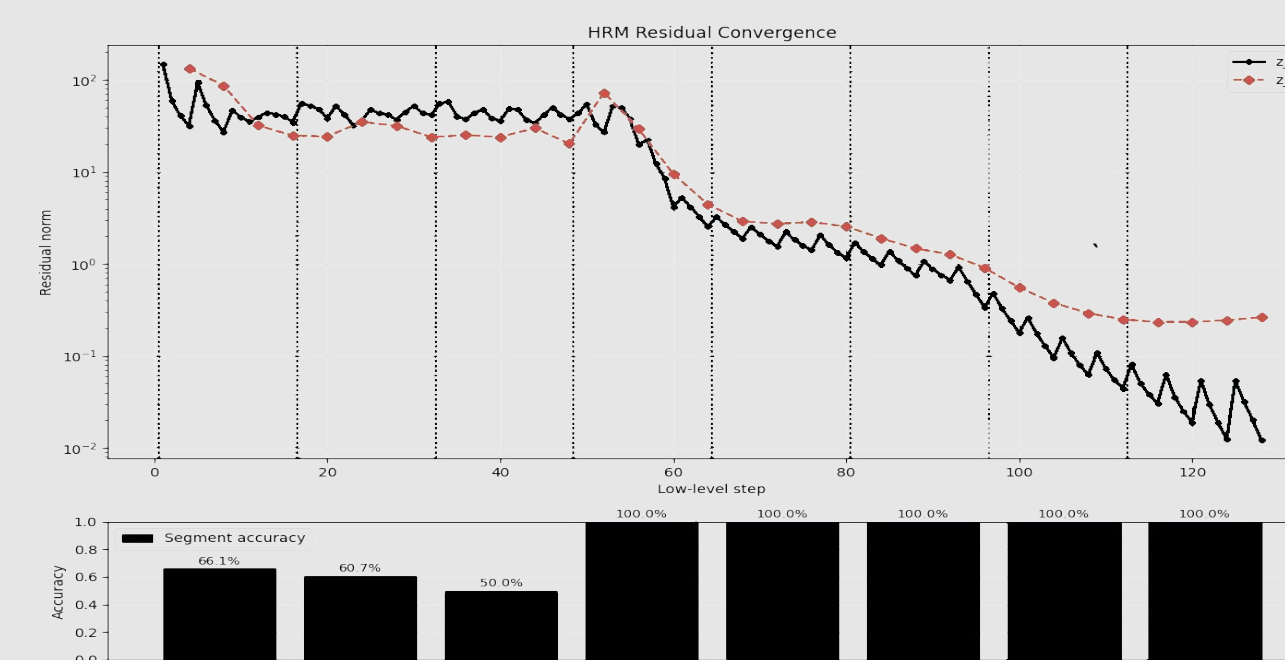
How does the model perform compared to baselines?

Convergence Compared to Encoder-Only Transformers and Bidirectional LSTMs



Are the paper's assumptions reasonable?

Exploring Hierarchical Convergence and Local Fixed Point Contraction



Local Contraction:

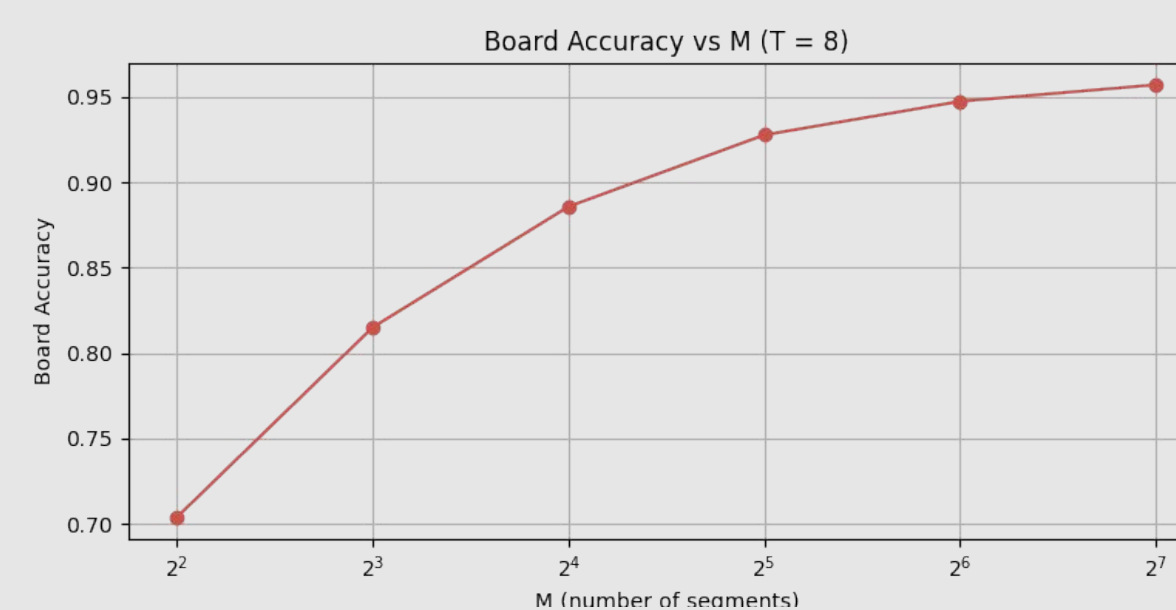
The model contracts the hidden state to the answer when it is within a small neighborhood / puzzle solved

Hierarchical Convergence:

The model stops changing the hidden state when it converges to the answer

Exploring the unaddressed implications of the model's assumptions

Adjusting Model Architecture Parameters After Training



The HRM's convergence pattern, with the shared parameter structure of the modules, implies

model size can be increased after training

so, we tried it out and found that **performance increases!**

04 Conclusion & Discussion

01 HRM converges to become a **extremely accurate model quickly**

02 Allows for **scaling without retraining** for improved accuracy

